Introduction to Cryptography

Lecture 15: Data Encryption Standard (DES)

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History of DES

- In 1971, IBM developed an encryption algorithm called **LUCIFER**.
- LUCIFER operated on 64-bit blocks.
- It used a 128-bit key.
- IBM researcher Walter Tuchman later refined the design.
- The key size was reduced to 56 bits.
- The refined algorithm became the basis for DES.

$$\boxed{\text{LUCIFER} \rightarrow \text{Refinement} \rightarrow \text{DES}}$$

DES as a Standard

DES was adopted in 1977 as the Data Encryption Standard.

- It became a widely used symmetric encryption algorithm.
- It was used in financial and commercial applications.
- DES was standardized as a federal encryption standard.

Important

DES encrypts data in **64-bit blocks** using an effective **56-bit key**.

Why Start with a Simplified DES Example?

The real DES algorithm processes:

64-bit blocks

and uses:

56-bit effective keys

This can be difficult to understand initially.

Therefore, we first consider a smaller DES-type system.

Simplified Example

Message size: 12 bits
Left half: 6 bits
Right half: 6 bits
Master key: 9 bits
Round key: 8 bits

Simplified DES-Type Algorithm

Suppose the 12-bit message is:

$$L_0 R_0$$

where:

$$L_0 = 6 \text{ bits}$$

and:

$$R_0 = 6 \text{ bits}$$

The key is:

$$K = 9 \text{ bits}$$

Each round uses:

$$K_i = 8 \text{ bits}$$

Simplified Feistel Round

The *i*th round transforms:

$$L_{i-1}R_{i-1}$$

into:

$$L_iR_i$$

using:

$$\boxed{L_i = R_{i-1}}$$

and:

$$\boxed{R_i = L_{i-1} \oplus f(R_{i-1}, K_i)}$$

This is the same fundamental Feistel idea used by DES.

Understanding the Feistel Equations

For every round:

$$L_i = R_{i-1}$$

This means:

The previous right half becomes the new left half.

Then:

$$R_i = L_{i-1} \oplus f(R_{i-1}, K_i)$$

This means:

The old left half is XORed with the output of the round function.

Simplified Round Function

Suppose the function:

$$f(R_{i-1}, K_i)$$

takes:

$$R_{i-1} = 6 \text{ bits}$$

and:

$$K_i = 8 \text{ bits}$$

The 6-bit right half is expanded into 8 bits.
Then:

$$E(R_{i-1}) \oplus K_i$$

is calculated.
The result is processed through simplified S-boxes.

Simplified Expansion Example

Suppose:

$$R_{i-1} = 011001$$

A simplified expansion may produce:

$$E(R_{i-1}) = 01010101$$

Thus:

$$6 \text{ bits} \rightarrow 8 \text{ bits}$$

The purpose is to make the data size equal to the round-key size.

Simplified XOR Example

Suppose:

$$E(R_{i-1}) = 10101010$$

and:

$$K_i = 01100101$$

Then:

$$10101010$$

$$\oplus$$

$$01100101$$

$$= \boxed{11001111}$$

Simplified S-Box Example

Suppose the 8-bit result is:

$$11001111$$

Divide it into two groups:

$$1100 \qquad 1111$$

Then:

$$S_1(1100) = 000$$

and:

$$S_2(1111) = 100$$

Therefore:

$$f(R_{i-1}, K_i) = \boxed{000100}$$

Simplified Round Numerical

Suppose:

$$L_{i-1} = 011100$$

$$R_{i-1} = 100110$$

and:

$$f(R_{i-1}, K_i) = 000100$$

Then:

$$L_i = R_{i-1}$$

$$L_i = 100110$$

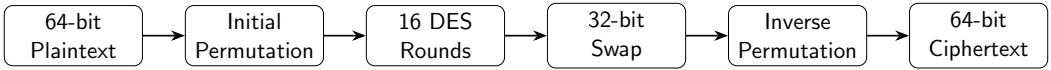
and:

Now Move to Real DES

Real DES uses the same basic Feistel principle.

Parameter	**DES**
Block size	64 bits
Effective key	56 bits
Number of rounds	16
Round-key size	48 bits
Left half	32 bits
Right half	32 bits
Number of S-boxes	8

Overall DES Encryption



Initial Permutation (IP)

The 64-bit plaintext first passes through:

$$\boxed{IP}$$

IP rearranges the bit positions according to a fixed table.
After IP:

$$IP(X) = L_0 R_0$$

where:

$$L_0 = 32 \text{ bits}$$

and:

$$R_0 = 32 \text{ bits}$$

Initial Permutation Table

58	50	42	34	26	18	10	2
60	52	44	36	28	20	12	4
62	54	46	38	30	22	14	6
64	56	48	40	32	24	16	8
57	49	41	33	25	17	9	1
59	51	43	35	27	19	11	3
61	53	45	37	29	21	13	5
63	55	47	39	31	23	15	7

For example, the first output bit is input bit 58.

Inverse Initial Permutation

After the sixteen rounds and final swap, DES applies:

$$\boxed{IP^{-1}}$$

The inverse permutation reverses the initial rearrangement.
Thus:

$$IP^{-1}(IP(X)) = X$$

and:

$$IP(IP^{-1}(X)) = X$$

DES Round Equations

For each round:

$$\boxed{L_i = R_{i-1}}$$

and:

$$\boxed{R_i = L_{i-1} \oplus F(R_{i-1}, K_i)}$$

The round key:

$$K_i$$

contains:

48 bits

There are:

16

DES Round Function

The function:

$$F(R, K)$$

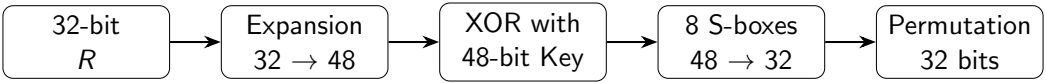
performs:

1. Expansion
2. XOR with round key
3. S-box substitution
4. Permutation

Thus:

$$\boxed{F(R, K) = P(S(E(R) \oplus K))}$$

Round Function Flow



Expansion Function

The right half:

$$R = 32 \text{ bits}$$

is expanded to:

$$48 \text{ bits}$$

The expansion table duplicates selected bits.

$$\boxed{32 \xrightarrow{E} 48}$$

The expanded value can then be XORed with the 48-bit round key.

DES Expansion Table

32	1	2	3	4	5
4	5	6	7	8	9
8	9	10	11	12	13
12	13	14	15	16	17
16	17	18	19	20	21
20	21	22	23	24	25
24	25	26	27	28	29
28	29	30	31	32	1

Notice that some bits appear more than once.

XOR with the Round Key

The expanded right half contains:

48 bits

The round key also contains:

48 bits

Thus:

$$B = E(R_{i-1}) \oplus K_i$$

The result is:

48 bits

which is then divided into eight 6-bit groups.

DES S-Boxes

DES contains:

$$\boxed{\text{8 S-boxes}}$$

The 48-bit input is divided into:

$$B_1, B_2, \ldots, B_8$$

Each block contains:

$$\text{6 bits}$$

Each S-box transforms:

$$\boxed{\text{6 bits} \rightarrow \text{4 bits}}$$

Therefore:

$$48 \rightarrow 32$$

S-Box Address Calculation

For a 6-bit input:

$$b_1 b_2 b_3 b_4 b_5 b_6$$

the row is determined by:

$$\boxed{b_1 b_6}$$

The column is determined by:

$$\boxed{b_2 b_3 b_4 b_5}$$

Thus:

$$\text{Row} = 2b_1 + b_6$$

and:

$$\text{Column} = 8b_2 + 4b_3 + 2b_4 + b_5$$

Actual DES S-Box Example

Consider the DES S_1 input:

$$011001$$

First and last bits:

$$01_2 = 1$$

Therefore:

$$\text{Row} = 1$$

Middle four bits:

$$1100_2 = 12$$

Therefore:

$$\text{Column} = 12$$

Permutation Function P

The eight S-boxes produce:

32 bits

These bits are rearranged using the fixed permutation P.

16	7	20	21	29	12	28	17
1	15	23	26	5	18	31	10
2	8	24	14	32	27	3	9
19	13	30	6	22	11	4	25

This helps spread S-box output bits across later rounds.

Complete DES Round

Remember the sequence:

$$\boxed{R \rightarrow E \rightarrow \oplus K \rightarrow S \rightarrow P}$$

Then:

$$L_i = R_{i-1}$$

and:

$$R_i = L_{i-1} \oplus F(R_{i-1}, K_i)$$

DES Key Generation

The original key is:

$$64 \text{ bits}$$

However:

$$8 \text{ bits}$$

are parity bits.
Therefore:

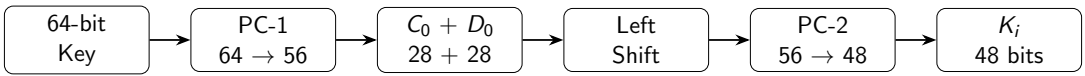
$$\boxed{64 - 8 = 56}$$

effective key bits remain.
DES generates:

$$\boxed{16 \text{ round keys}}$$

each containing:

Key Generation Process



Permuted Choice 1

PC-1 takes:

$$\text{64-bit key}$$

and produces:

$$\text{56 bits}$$

The result is divided into:

$$C_0 = 28 \text{ bits}$$

and:

$$D_0 = 28 \text{ bits}$$

Thus:

$$\boxed{C_0 D_0 = 56 \text{ bits}}$$

Left Shift Schedule

Round	1	2	3	4	5	6	7	8	9	10	11	12	13	14	15	16
Shift	1	1	2	2	2	2	2	2	1	2	2	2	2	2	2	1

For every round, both:

$$C_i$$

and:

$$D_i$$

are circularly shifted left.

Permuted Choice 2

After shifting:

$$C_i D_i$$

contains:

56 bits

PC-2 selects and rearranges:

48 bits

Therefore:

$$\boxed{K_i = PC2(C_i D_i)}$$

The result is the round key.

DES Decryption

DES decryption uses the same basic algorithm as encryption.
The only major difference is the order of the keys.
Encryption:

$$K_1, K_2, \ldots, K_{16}$$

Decryption:

$$\boxed{K_{16}, K_{15}, \ldots, K_1}$$

This is an important advantage of the Feistel structure.

Strength of DES

When DES was introduced:

- 2^{56} possible keys represented a large key space.
- DES had a strong internal structure.
- Its S-boxes provided nonlinear substitution.
- Sixteen rounds provided substantial confusion and diffusion.

However, concerns were raised about:

- The reduced key size.
- The secrecy of the original S-box design criteria.

DES Key Space

DES uses:

56-bit effective key

Therefore:

$$\boxed{2^{56}}$$

possible keys exist.
Numerically:

$$2^{56} \approx 7.2 \times 10^{16}$$

This was once considered extremely large.
Today, it is insufficient.

Avalanche Effect

A desirable property of encryption is:

Avalanche Effect

A small change in the plaintext or key should cause a large change in the ciphertext.
For example:

Change 1 plaintext bit

should cause:

many ciphertext bits to change

DES exhibits a strong avalanche effect after only a few rounds.

Avalanche Effect in DES

In a standard DES example, changing one plaintext bit causes progressively more differences across the rounds.
After several rounds, many bits differ.
By the completion of DES, approximately half of the ciphertext bits may differ.

$$\boxed{\text{Small Input Change} \rightarrow \text{Large Output Change}}$$

Why DES Became Insecure

The main practical weakness of DES is:

56-bit key size

As computing power increased, brute-force attacks became practical.
Historical demonstrations showed that DES could eventually be broken using specialized hardware and distributed computing.

Therefore:

DES is obsolete for new secure systems

DES Evolution

DES was followed by:

Triple DES

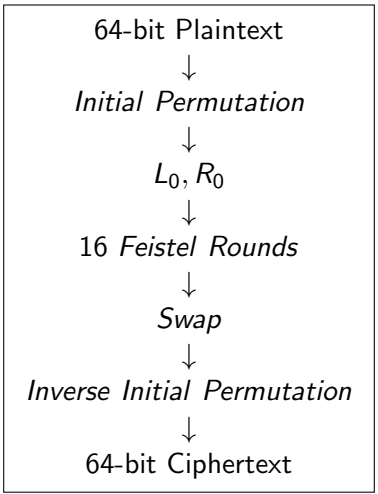
and later:

AES

$$\text{DES} \rightarrow \text{3DES} \rightarrow \text{AES}$$

AES is the preferred modern symmetric block cipher.

DES: Complete Revision



Round Function Revision

$$\boxed{F(R, K) = P(S(E(R) \oplus K))}$$

Remember:

$$\boxed{\text{Expand} \rightarrow \text{XOR} \rightarrow \text{Substitute} \rightarrow \text{Permute}}$$

Quick Classroom Questions

1. What is the block size of DES?
2. What is the effective DES key size?
3. Why does DES use only 56 effective key bits?
4. How many rounds are used?
5. What is the size of each round key?
6. Why is the right half expanded?
7. What is the role of the S-box?
8. What is the avalanche effect?
9. How is DES decryption different from encryption?
10. Why is DES no longer considered secure?

References

1. William Stallings, *Cryptography and Network Security: Principles and Practice*, Pearson.
2. NIST, *Data Encryption Standard (DES)*, FIPS Publication 46.
3. Course reference presentation: *Data Encryption Standard (DES)*.
4. Alfred J. Menezes, Paul C. van Oorschot, Scott A. Vanstone, *Handbook of Applied Cryptography*.

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TCS-392 – Introduction to Cryptography
Lecture 15 – DES Algorithm

Thank You

Questions?